COMMUNICATIONS 2025-2026'S 1ST SEMSTER MID EXAM:

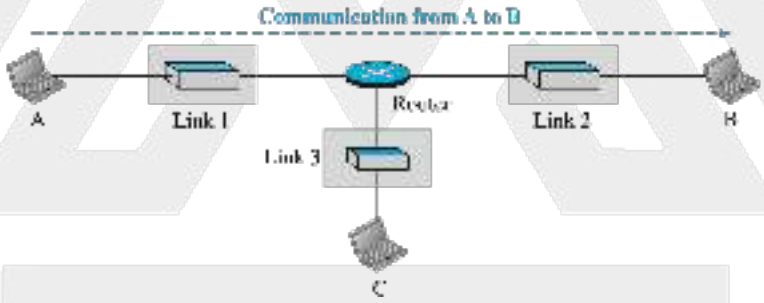
Question No. 1 (24 Marks):

Q1. Mesh topology with 4 stations, how many connections in this topology?
- A) 6
- B) 12
- C) 3

Q2. The TCP/IP suite is:
- A) unidirectional communication system
- B) bidirectional communication system
- C) multidirectional communication system

Q3. The TCP/IP suite is:
- A) software layers
- B) software and hardware layers
- C) hardware layers

For the following figure "Answer Q.4-Q.5":



Q4. The router is involved in:
- A) all five layers
- B) only three layers
- C) only two layers

Q5. The station that makes no encapsulation/decapsulation is:
- A) sender
- B) router
- C) link

Q6. The addresses of the source & destination hosts are added as a header to the message.
- A) network layer
- B) data-link layer
- C) transport layer

Q7. Changing TV channels by IR remote control is:
- A) point-to-point half-duplex communication
- B) point-to-point simplex communication
- C) point-to-point full-duplex communication

Q8. If the network stores and forwards a packet, then this network is called:
A) switched network
B) circuit-switched network
C) packet-switched network

Q9. DSL service is an upgraded service with respect to:
A) Dial-up service
B) Wi-Fi
C) ISP

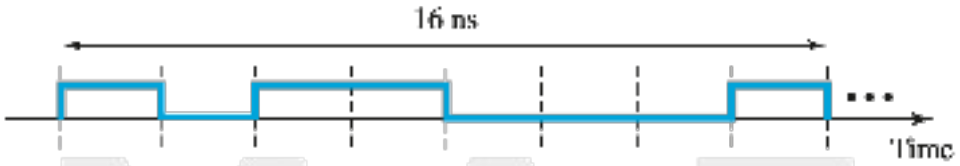
Q10. Variation in the packet arrival time is called:
A) timeliness
B) jitter
C) timing

Q11. The bandwidth of a TV channel is usually measured in:
A) KHz
B) GHz
C) MHz

Q12. The channel in broadband transmission is called:
A) low-pass channel
B) band-pass channel
C) pass-band channel

Q13. The spectrum of voice signals is:
A) discrete
B) continuous
C) non-periodic

Q14. The bit rate of the following signal is:



A) 500 mbps
B) 500 μbps
C) 500 Mbps

Q15. We measure the performance of a telephone line (4 kHz of bandwidth). When the signal is 10 V, the noise is 5 mV. What is the maximum data rate supported by this telephone line?
A) 43866 bps
B) 43863 bps
C) 43.866 bps

Q16. What is the phase shift of a sin wave with maximum amplitude after 1/4 cycle?
A) 90 degrees
B) 0 degrees
C) -90 degrees

Q	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
Ans	B	B	A	B	C	A	B	C	A	B	C	B	B	C	B	C

Question No. 2 (6 Marks):

What is the length of a bit in a channel with a propagation speed of 2×10^8 m/s if the channel bandwidth:
A) 1 Mbps
B) 10 Mbps
C) 100 Mbps

Solution:

$$Bit\ Length = Propagation\ Speed \cdot Time$$

A)

$$Bit\ Length = 2 \times 10^8 \text{ m/s} \cdot \frac{1}{1 \times 10^6 \text{ bps}} = 200 \text{ m for each bit}$$

B) 10 Mbps

$$Bit\ Length = 2 \times 10^8 \text{ m/s} \cdot \frac{1}{10 \times 10^6 \text{ bps}} = 20 \text{ m for each bit}$$

C) 100 Mbps

$$Bit\ Length = 2 \times 10^8 \text{ m/s} \cdot \frac{1}{100 \times 10^6 \text{ bps}} = 2 \text{ m for each bit}$$

References Pictures

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Q1:(Please write your answers into table only, otherwise your answers will be neglected) **(24 marks)**

Question	1	2	3	4	5	6	7	8	9	10
Answer	a	b	b	b	a	a	b	c	a	a
Question	11	12	13	14	15	16				
Answer	c	a	b	c	d	a				

+12

1. Mesh topology with 4 stations, how many connections in this topology:

a) 6 b) 12 c) 3



2. The TCP/IP suite is: 5layers

a) unidirectional communication system
b) bidirectional communication system
c) multidirectional communication system

3. The TCP/IP suite is:

a) software layers
b) software and hardware layers
c) hardware layers

4. The router in the following figure is involved in:



Network
link
Physical

a) all five layers b) only three layers c) only two layers

5. Due to the following figure, the station that makes no encapsulation/decapsulation is:

a) sender b) router c) link

Both

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6. The addresses of the source & destination hosts are added as a header to the message in:
a) network layer b) data-link layer c) transport layer

7. Changing TV channels by IR remote control is:
a) point-to-point half-duplex communication
b) point-to-point simplex communication
c) point-to-point full-duplex communication

8. If the network stores and forwards a packet, then this network is called:
a) switched network
b) circuit-switched network
c) packet-switched network

9. DSL service is an upgraded service with respect to:
a) dial-up service b) Wi-Fi c) ISP

10. Variation in the packet arrival time is called:
a) timeliness b) jitter c) timing

11. The bandwidth of a TV channel is usually measured in:
a) kHz b) GHz c) MHz

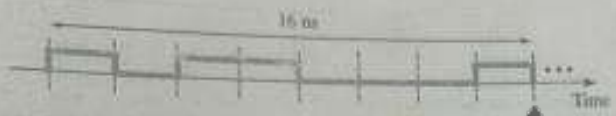
12. The channel in broadband transmission is called:
a) low-pass channel
b) band-pass channel
c) pass-band channel

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13. The spectrum of voice signals is:
a) discrete (b) continuous c) non-periodic

14. The bit rate of the following signal is:



a) 500 mbps b) 500 µbps (c) 500 Mbps

15. We measure the performance of a telephone line (4 KHz of bandwidth). When the signal is 10 V; the noise is 5 mV. What is the maximum data rate supported by this telephone line?
a) 43866 bps b) 43863 bps c) 43.866 bps

16. What is the phase shift of a sin wave with maximum amplitude after 1/4 cycle:
(a) 90 degrees b) 0 degrees c) -90 degrees

Q2: What is the length of a bit in a channel with a propagation speed of 2×10^8 m/s if the channel bandwidth is
a. 1 Mbps?
b. 10 Mbps?
c. 100 Mbps?
(6 marks)

(*) Bit Length (m) = Prop. speed (m/s) * time (s)

a) Bit length = $2*10^8 * \frac{1}{1\,M} \Rightarrow 2*10^8 * 1*10^{-6} = 200$ m

b) Bit length = $2*10^8 * \frac{1}{10\,M} \Rightarrow 2*10^8 * 10*10^{-6} = 20$ m

c) Bit Length = $2*10^8 * \frac{1}{100\,M} \Rightarrow 2*10^8 * 100*10^{-6} = 2$ m

GOOD LUCK